

Reproducibility checklist (Complete and submit as a standalone file with your project report)

- * Please fill out the top part of the form first (first page).
- * Please make sure the points in second page are addressed in your report submission
- * Please copy this and replace the ☐ with a ☒ for the items that are addressed in your report/code submission
- * Please complete this checklist, attach it to your final project report as the last page and then submit.

Your team member names (comma separated):

Lixing Lin, Youxuan Ma, Peng Chen

Your project title:

Reasoning Trace Topology as Calibration-Free Uncertainty Quantification for Large Language Models

AI Use Statement (see below for instructions and examples)

We used Anthropic's Claude (Sonnet 4.6 / Opus 4.6) as a coding and writing assistant during this project. It helped us with code review and debugging, as well as some statistical analysis. We also used it to review our report and incorporated its suggestions to polish and enhance our final draft. In addition, we used it as an idea sounding-board and discussed with it whether to drop or retain individual signal families and what ablations would be informative.

Author Contribution Statement (see below for instructions and examples)

Youxuan Ma handled all datasets ingestion and preparation, including trace generation, data preprocessing, integration, and feature engineering. Contributed to the architectural design and model development, and helped improve final model performance through several optimizations including multi-layer probes. Organized and maintained the code repo, and led the writing of the final report. Lixing Lin implemented the sequence-model baselines (DeBERTa, Longformer-base-4096), developed the step-embedding transformer, designed the weighted ensemble and hybrid meta-learner stacking pipeline, ran the main fine-tuning experiments on the Yale Bouchet HPC cluster, and optimized the training pipeline (pre-tokenization, focal loss, early stopping). Peng Chen designed the structure-aware feature pipeline, including a rule-based six-class behavior parser, 25

handcrafted descriptors (length, structural, meta), 5 semantic-recurrence features, 4 problem-trace alignment features, and 10 generation-uncertainty features from per-token logprob and entropy trajectories. Built the length-controlled evaluation framework. Designed and ran the hidden-state probing experiments, including the layer-position atlas sweep (8 layers, 4 positions, Qwen and Llama) and the multi-layer concatenation probe. Implemented the conformal-prediction wrapper with cross-family and leave-one-dataset-out coverage validation, and the per-dataset and cross-model transfer evaluation harness. Diagnosed and fixed a cartesian-product merge bug in the meta-learner that had inflated reported AUROC by over 0.10. Contributed to result analysis in the report.

Checklist (Make sure these are addressed in your submission and replace ☐ with a ☒ where applies)

Source Code Accessibility:

- ✓ Provide a link to the source code on github or submit the code repo to us separately if you have privacy concerns.
- ✓ Ensure the code is well-documented and easy to run

Computing Infrastructure:

- ✓ Mention the compute resources used (GPUs or API models you've used)

Dataset Description:

- ✓ Clearly describe and cite the datasets used, and any preprocessing/modifications done.

Hyperparameters and Tuning Process:

- ✓ Detail the hyperparameters used and the process for selecting them.
(Example: The model was fine-tuned using a batch size of 16, learning rate of 1e-5, and trained on 1000 steps with 100 steps of learning rate linear warmup with linear decay)

Experimental Results:

- ✓ Clearly define the evaluation metrics and statistical methods used in assessing the model.
- ✓ Present the main set of results in a table, chart or readable format
- ✓ Include comparisons with baseline models and state-of-the-art, where applicable.

AI Use

- ✓ Clear statement of how and where you have used AI in the project
- ✓ Specific parts where AI has helped
- ✓ Prompts that you used included as an auxiliary file or appendix in your report

Good Example 1: We used ChatGPT to help brainstorm project directions, clarify implementation details. We then used Claude Code to help write code and experiments. We used both ChatGPT and Claude Code to help plot bar charts for our raw results. All experimental design decisions, code changes, analysis, and final writing were reviewed and finalized by the team. We used ChatGPT to help with polishing the report write-up and help with grammar.

Good Example 2: We used Codex to assist with implementing preprocessing scripts for the summarization dataset, including tokenization, train/dev/test splitting, and formatting examples for model input. We also used Claude Code to help debug errors in the fine-tuning script and to refactor evaluation code for ROUGE scoring. All prompts, generated code, and edits were reviewed by the team before inclusion in the final repository. We did NOT use AI for the final project report writing extensively with exception of helping rephrasing a few sentences.

Bad example 1: We vibe-coded parts of the project using Claude Code.

Bad example 2: AI was used in Writing, Coding and Experiments.
(This doesn't say how exactly you used AI and in what specific parts)

Author Contribution Statement

✓ *Clear statement of what each member in your team has done.*

Good example 1: Alice Smith led dataset collection and preprocessing, including cleaning the raw data and creating the train, validation, and test splits. Bob Chen implemented the baseline and fine-tuned models, ran the main experiments, and organized the code repository. Carol Lee designed the evaluation framework, produced the result tables and figures, and wrote the analysis section of the report. All team members contributed to project planning, reviewed the final report, and approved the final submission.

Good example 2 (mix of grad and undergrad students): Bob Chen (UG) handled dataset preprocessing and exploratory analysis. Carol Lee (UG) implemented the model training pipeline and ran the hyperparameter tuning experiments. Alice Smith (Grad) conducted evaluation, created the result visualizations, and wrote the discussion of errors and limitations. Alice also conducted the literature review. All authors contributed to writing and editing the final report.

Bad example 1: All team members contributed equally to the project.
(This gives no information about each person's role or contribution.)

Bad example 2: Alice did coding, Bob did experiments, and Carol wrote the report.
(This is too general, you need to provide specific details on exactly who did what)